

KOHLER AI · CONCEPT PROTOTYPE

The Dream Bathroom Studio

An AI-powered, immersive bathroom design platform that turns a customer's dimensions, preferences, and budget into a personalized, spatially validated design.

Dream



Discover



Design



Refine



Visualize



Finalize

github.com/atharv2609/dream-bathroom-design

A personal interior designer, not another product catalog

- Feels like a luxury home-design studio, not a generic e-commerce site or static chatbot.
- Converts room dimensions, layout, preferences and budget into a personalized KOHLER-style design.
- Combines editorial-style visual inspiration with the intelligence of a professional bathroom planner.

THREE CONNECTED EXPERIENCES

A

Dream Studio

Immersive landing page, inspiration gallery, clickable bathroom styles.

B

Design Studio

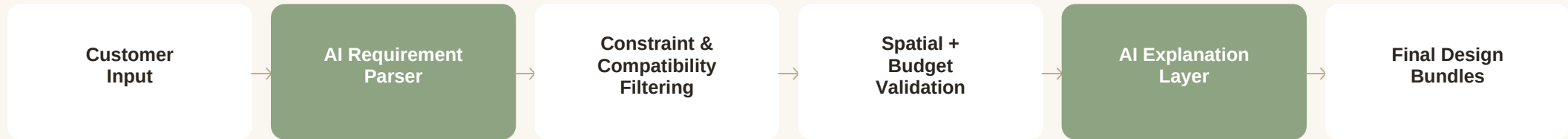
Guided wizard — requirements, recommendations, 2D floor plan, budget, AI assistant.

C

Design Results

Final proposal — bundle, layout, rationale, sustainability, export.

Deterministic logic decides. AI explains and converses.



Hard constraints (dimensions, clearances, compatibility, budget) are never overridden by the LLM. Soft scoring ranks feasible options on spatial fit, budget fit, style match, compatibility, water efficiency and feature match — weights are configurable, and scores are shown as decision support, never as guarantees.

Interactive 2D / optional 3D layout

SVG/Canvas floor plan validates overlaps, clearances and door-swing collisions in real time; stays synced to the selected bundle.

Conversational refinement

“Make it more luxurious” or “reduce cost by Rs 50,000” re-ranks and recalculates while preserving mandatory requirements.

Built to be trustworthy, not just impressive

TECH STACK

- React + TypeScript + Vite, Tailwind CSS
- SVG/Canvas 2D planner; optional Three.js / React Three Fiber 3D
- Node.js with Express or Next.js API routes
- SQLite / PostgreSQL / Supabase data layer
- Provider-independent LLM abstraction with deterministic fallback

RESPONSIBLE BY DESIGN

- Synthetic catalog clearly labeled; no fabricated KOHLER claims or reviews
- No facial recognition; consent required before processing uploaded images
- AI-derived measurements always flagged as estimates, not verified data
- WCAG 2.2 AA target: keyboard nav, contrast, reduced motion

Status

Scaffolded with Lovable — React/TypeScript/Vite/Tailwind structure in place.
Recommendation engine, spatial validator and AI layer per spec: in progress.